SS1 PHYSICS LESSON NOTES (THIRD TERM)
:

WEEK 6 – PROPERTIES OF WAVES

PROPERTIES OF WAVES

(i) Reflection

Reflection is the bouncing back of a wave when it hits a surface.

Example: light reflecting from a mirror.

(ii) Refraction

Refraction is the bending of a wave as it passes from one medium to another due to change in speed.

(iii) Diffraction

Diffraction is the spreading of waves as they pass through a small opening or around an obstacle.

(iv) Interference

Interference occurs when two or more waves overlap to form a new wave pattern.

Types:

- Constructive interference (increase in amplitude)
- Destructive interference (decrease in amplitude)

(v) Stationary Waves

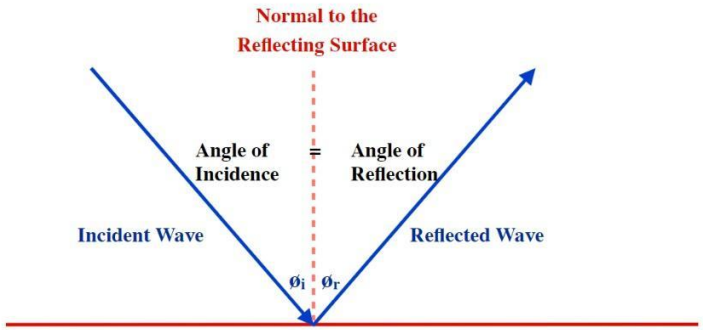
Stationary waves are formed when two waves of equal amplitude and frequency move in opposite directions and combine.

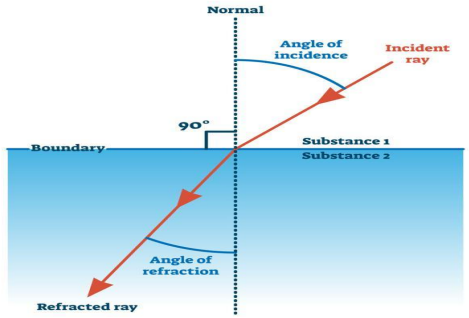
(vi) Polarization

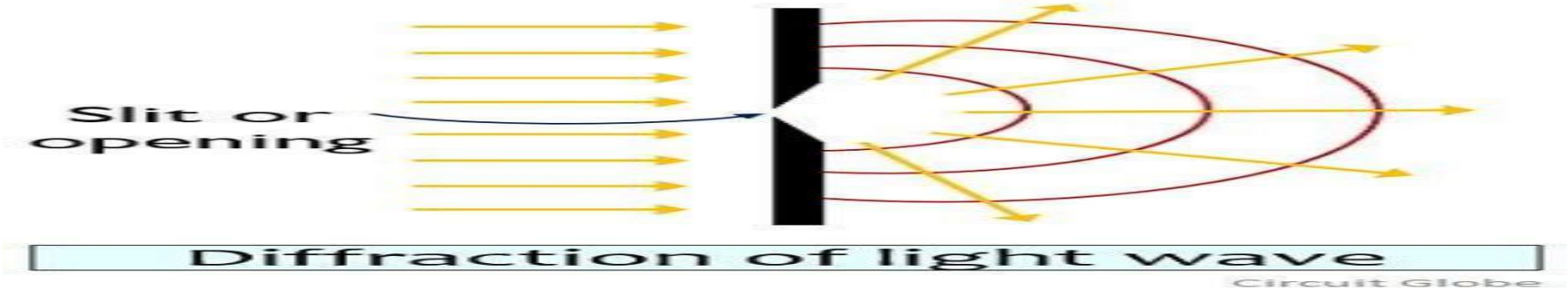
Polarization is the restriction of wave vibrations to one plane.

VISUAL REPRESENTATION OF WAVE PROPERTIES

REFRACTION







WAVE EQUATION

$Y = A \sin(\omega t - (2\pi x / \lambda))$

Where:

- A = amplitude
- ω = angular frequency
- λ = wavelength
- t = time
- x = position

GRAPHICAL REPRESENTATION OF WAVES

Waves can be represented in two ways:

- Displacement vs distance
- Displacement vs time

These show amplitude, wavelength, and frequency.

WAVE SPEED FORMULA

$v = f \times \lambda$

Where:

- v = wave speed (m/s)
- f = frequency (Hz)
- λ = wavelength (m)

WORKED EXAMPLES

Example 1

A wave travels from deep water to shallow water.

Given:

- $v_1 = 15$ m/s
- $\theta_1 = 45°$
- $\theta_2 = 30°$

Using Snell's law for waves:

$\sin \theta_1 / \sin \theta_2 = v_1 / v_2$

Substitute:

$\sin 45° / \sin 30° = 15 / v_2$

$(0.707 / 0.5) = 15 / v_2$

$1.414 = 15 / v_2$

$v_2 = 15 / 1.414$

$v_2 = 10.6$ m/s

✔ Final Answer: 10.6 m/s

Example 2

Radio waves in air:

$v = 3.0 \times 10^8$ m/s
Refractive index of water = 4/3

Speed in water:

$v_2 = v / n$

$v_2 = (3 \times 10^8) / (4/3)$

$v_2 = 3 \times 10^8 \times 3/4$

$v_2 = 2.25 \times 10^8$ m/s

Example 3

Given:

- refractive index = 1.5
- angle of incidence = 60°

Using:

$n = \sin i / \sin r$

$1.5 = \sin 60° / \sin r$

$\sin r = \sin 60° / 1.5$

$\sin r = 0.866 / 1.5$

$\sin r = 0.577$

$r = \sin^{-1}(0.577)$

$r = 35°$

✔ Final Answer: 35°

Example 4

Given:

- $v = 3 \times 10^8$ m/s
- $\lambda = 150$ m

$f = v / \lambda$

$f = (3 \times 10^8) / 150$

$f = 2 \times 10^6$ Hz

✔ Final Answer: 2.0×10^6 Hz

EVALUATION QUESTIONS

1. Define:
    - Amplitude
    - Wavelength

1.
    1. Frequency
    2. Period
    3. Wave speed
2. Draw and label:
    1. Transverse wave
    2. Longitudinal wave
3. A wave travels at 340 m/s with frequency 170 Hz. Find wavelength.

$\lambda = v / f$
$\lambda = 340 / 170$
$\lambda = 2$ m

✔ Answer: 2 m

ASSIGNMENT

1. Explain reflection, refraction, diffraction, and interference
2. Draw:

1. Wave reflection
2. Interference pattern
3. Diffraction through slit

3. Solve wave problems using $v = f\lambda$